Renaldo Miya

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EDUCATION

University of Waterloo **Expected Graduation: April 2029**
BASc, Honours Mechanical Engineering

WORK EXPERIENCE

Design Engineering Intern **September 2026 – December 2026**
Amphenol Markham, ON

- Redesigned an RJ45 housing snap-fit latch to exceed a competitor's retention force, reworking beam geometry to cut peak root stress 25%. Validated hand calcs against SolidWorks Simulation FEA and released to production
- Designed an IEC 60999-1 test fixture to measure plug contact resistance under dynamic motion as a lower-cost alternative to a $2K commercial rig; secured management approval and coordinated fabrication
- Released GD&T detail and manufacturing drawings for 10+ connector and receptacle part numbers
- Built SolidWorks drawing templates that cut update time ~30% across 5 product lines
- Implemented ECNs in Windchill by updating part models and revising customer-facing sales drawings

Mechanical Engineering Assistant **January 2026 – April 2026**
TRIUMF | Canada's national particle accelerator laboratory Vancouver, BC

- Owned the design of a wall-mounted stainless support frame for ARIEL's radioactive-gas manifold, from requirements (ASME B31.3 practice, design for radiation) to fabrication-ready drawings
- Cut peak deflection from 6.5 mm to 1.1 mm and peak stress from 107 MPa to 50 MPa by optimizing clamp layout and member sizing in ANSYS, while reducing part count 25% to shorten installation time and dose exposure
- Released ASME Y14.5 GD&T drawings and managed BOMs and revisions in SolidWorks PDM; modelled beamline shielding components and revised them through design reviews with senior engineers
- Machined experiment components on CNC, manual mills, and lathes

Mechanical Engineering Co-op **May 2024 – August 2024**
BC Hydro | Site C Clean Energy Project Fort St John, BC

- Reviewed and redlined P&IDs and mechanical layouts against applicable codes and standards, flagging constructability issues and issuing modifications adopted into the build
- Conducted on-site field inspections to verify installation accuracy and code compliance, documenting discrepancies and driving corrective design changes that reduced rework

PROJECTS & DESIGN TEAMS

Force-Feedback Robotic Gripper | SolidWorks, Arduino Uno (C++), 3D Printing **Fall 2026 – Present**

- Designed a two-finger, servo-actuated rack and pinion gripper in SolidWorks for FDM printing, using a PLA frame with flexible TPU finger pads for better grip
- Integrated FSR 402 force sensors with an Arduino Uno to run a closed-loop proportional controller that adjusts servo position to hold a target grip force, handling fragile and rigid objects without crushing them
- Calculated gear module and tooth profiles to minimize backlash within the rack and pinion mechanism, ensuring precise positioning of the 3D-printed PLA/TPU fingers

UW Formula Electric | Mechanical Design & Composites Member **2025**

- Designed a 3D-printed enclosure for the vehicle's power distribution relays in SolidWorks, cutting mass 20% from the first iteration and reaching a flush fit with fixed chassis mounts and connectors in 3 printed iterations
- Built composite parts by hand, moulding the undertray and laying up carbon fibre plating, and machined components on shop equipment

TECHNICAL SKILLS

CAD & Modelling | SolidWorks, AutoCAD, DraftSight
Analysis & Simulation | ANSYS (Mechanical, Fluent), SolidWorks Simulation, MATLAB
Prototyping & Fabrication| 3D Printing, Silicone Moulding, CNC Machining, Mill, Lathe, Drill Press, Composite Layup
Design & Documentation | GD&T (ASME Y14.5), DFM/DFA, Fixture Design, PDM, Windchill
Programming | C++, C, Python, HTML/CSS